

AI AND MACHINE LEARNING

ITI's AI & Machine Learning Track develops highly qualified AI engineers through 1065 hours of comprehensive, industry-aligned training. Program graduates demonstrate proven competencies in machine learning, deep learning, computer vision, natural language processing, and generative AI — supported by practical experience in MLOps, big data technologies, and cloud infrastructure. Organizations partnering with ITI gain access to a pipeline of well-prepared professionals capable of delivering impactful AI solutions from the outset.

JOB PROFILES

**Agentic AI Engineer · Artificial Intelligence Engineer ·
Generative AI Engineer · Natural Language Processing (NLP) Engineer ·
Computer Vision Engineer · MLOps Engineer · LLM Engineer ·
Machine Learning Engineer · Data Scientist**



TOTAL HOURS

1,329^{hrs}


DELIVERY

Blended


UNIVERSITY GRADE

Good


CERTIFICATIONS

4


Where graduates work

VOIS

Vodafone Egypt

Orange

Etisalat &

Bank Misr

National Bank of Egypt

PWC

IBM

Aman

Deloitte

Ejada

Microsoft

Data Science Middle East



Certifications earned

MLS-C01 — AWS Certified Machine Learning - Specialty

AIP-C01 — AWS Certified Generative AI Developer - Professional

AI-901 — Microsoft Azure AI Fundamentals

AI-300 — Machine Learning Operations Engineer Associate



Selection process

01 IQ & English Exam



Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam



Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview



Interview with the technical committee.

04 Interpersonal Interview



Candidates who pass the technical interview are invited to this final stage.



Who may apply

Graduates of:

- o Engineering
- o Science (Computer Science, Statistics and Mathematics)
- o Computing and Informatics
- o Computing and Artificial Intelligence
- o Computing and Data Science

Program content – 1,329 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● General Foundation Courses	186 hrs	14%	Agile Software Development Methodologies, Introduction to Programming using C++, Data structures & Algorithms, Linux Operating System, Version Control (GIT), Database Fundamentals using SQL
● Core Courses	687 hrs	52%	Introduction to AI, Python for Machine Learning, Linear Algebra For Machine Learning, Probability & Statistics For Machine Learning, Numerical Optimization For Machine Learning, Data Exploration Preparation and Visualization, Machine Learning I : Supervised ML, Machine Learning II, Bayesian and Unsupervised, Machine Learning III: Generative AI and prompt Engineering, Machine Learning IV: Agentic AI, Deep Learning, Advanced Computer Vision, Natural Language Processing and Large Language Models, Time Series Analysis and Forecasting, Recommender Systems, Machine Learning in Production (MLOps), Web Development using Django & FastAPI, Cloud Infrastructure for Big Data, Hadoop, Hive and Kafka, Spark & Pyspark for Big Data, Database Fundamentals using NoSQL
● English & Life Skills Courses	156 hrs	12%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor (Workshop), English
● Graduation Project	300 hrs	23%	Graduation Project

Student Deliverable

- Graduation project
- Freelance task (optional)
- Certification (optional)

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

[Probability & Statistics for Machine Learning & Data Science](https://www.coursera.org/learn/machine-learning-probability-and-statistics)
https://www.coursera.org/learn/machine-learning-probability-and-statistics

[Mathematics \(Calculus and Linear Algebra\)](https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/)
https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/

[CS50's Introduction to Computer Science](https://cs50.harvard.edu/x/)
https://cs50.harvard.edu/x/

[Database and SQL](https://maharatech.gov.eg/course/view.php?id=740)
https://maharatech.gov.eg/course/view.php?id=740

[ITI values](https://iti.gov.eg/about-us)
https://iti.gov.eg/about-us

APPLY AT

ITI.GOV.EG